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Dynamic Tracking: A Machine Learning Approach for PV Yield Optimization

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ABSTRACT

This work presents a novel framework for photovoltaic (PV) tracking optimization called dynamic tracking. Its core is a method of irradiance prediction using a machine learning surrogate model trained on ray tracer simulations. This approach outperforms ray tracing models in terms of computational speed while upholding accuracy for diffuse and direct irradiance. It is therefore able to predict irradiance distributions with a resolution of $11 \times 11 \text{ cm}^2$ for all possible tilting configurations over a given weather time series in real time. Spatial and temporal validation under overcast and sunny conditions show that the network accurately reproduces direct and diffuse shading patterns under arbitrary weather conditions. The proposed framework simulates all possible tilt sequences for any single-axis tracking system over daily periods with a 10 min resolution in only 2 min. Using discrete dynamic programming, it then calculates the truly optimal tilting sequence with respect to average irradiation. This method predicts and avoids inter-row shading, thereby increasing yield and outperforming established strategies like backtracking or diffuse tracking methods for any weather condition.

1 | Introduction

Modern photovoltaic (PV) power plants widely use axial tracking systems to optimize plane of array irradiation and land usage [1, 2]. The amount of usable solar irradiance at a given site fluctuates with time of day, season, and atmospheric conditions. While being simple and cost-effective, fixed-tilt arrays cannot adapt to these temporal and spatial variations in sunlight. Tracking systems overcome this limitation to maximize PV yield by dynamically reorienting modules to follow the sun's motion over the course of a day.

In recent years, different tracking strategies have evolved to tackle the challenges of tracking to maximize irradiance and avoid self-shading [1–3]. The simplest tracking strategy aims to tilt the PV modules to minimize the angle of incidence, thereby

maximizing incoming irradiance. For multiple rows of PV modules, this method is prone to inter-row shading, especially for times with a low apparent sun altitude. As PV modules consist of substrings connected in series, cells under shading can create a current bottleneck for the current flowing through multiple cells, so avoidance of self-shading is crucial. A common strategy to mitigate this problem is called backtracking, which limits the tracking angle to avoid self-shading from direct normal irradiation (DNI) under clear-sky conditions, leading to a more homogeneous light distribution on the PV string.

Further challenges and tracking strategies arise in the case of diffuse and mixed weather conditions. Under overcast skies, the light distribution behaves drastically differently, leading to the need for adapted tracking strategies. Daylight under cloud coverage mainly consists of diffuse horizontal irradiation (DHI),

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which is solar radiation that has been scattered in the atmosphere and arrives at the surface from a wide range of angles, rather than directly from the solar disk. As the high DHI to DNI ratio can lead to power loss due to diffuse inter-row shading [4, 5], modules are driven toward horizontal tilts to maximize diffuse capture and reduce these effects. Tracking strategies also considering diffuse irradiation are often referred to as diffuse tracking strategies. Often, mixtures of diffuse tracking strategies and backtracking are employed for scenarios with mixed irradiation [6, 7].

Optimal mixed-weather tracking strategies are under investigation and a solution is presented within this work. As many strategies rely on distinct empirically adjusted weather scenario cases [6, 7], it is common to switch between backtracking under clear skies and horizontal tilts under partial or full cloud coverage. These cases are usually distinguished by thresholding of weather parameters, which inherently exhibits occasional abrupt transitions between distinct tracking modes. The overlap between these cases is often empirically found. Optimal strategies for varying or mixed-weather conditions are therefore still to be determined. Our proposed tracking method tries to mitigate these effects by simulating multiple timesteps at once. It thereby prevents abrupt switching between strategies.

Ray tracing simulations are very accurate in simulating light behavior but always pose a tradeoff between accuracy and computation time. By simulating a vast number of rays individually, ray tracing can effectively simulate the effects of shading even under diffuse light [8–10]. In recent years, Deline et al. [8], used ray tracing to simulate the mismatch loss of bifacial PV modules for different albedos. In our previous work, Bruno et al. [11], applied ray tracing in the agri-PV context to develop tracking strategies considering the irradiance distribution on crops beneath the PV array. Due to the computational cost, the use of ray tracing in real-time optimization is limited. To achieve the precision needed to simulate diffuse light scattering, thousands of rays must be calculated.

A way to mitigate the downsides of ray tracing is by using machine learning. Many works use machine learning techniques to predict irradiance in PV scenarios [12–14]. These usually

predict statistical values, focusing on forecasting for larger regions by weather data only. This work follows the approach from our previous research published by Kurumundayil et al. [15], who developed a surrogate model to predict shadows of PV modules with a high spatial resolution. By encoding the PV scene, their deep neural network learns to predict irradiance maps on ground level based on simulated ray tracing data with sensors placed beneath the modules. Thereby, they circumvent the computation time of ray tracing while maintaining its accuracy. Our work will build on this technique to accurately predict the effect of different tilt and weather configurations on PV module irradiance in real time.

We present a novel method for tracking optimization, which we call dynamic tracking. It allows for real-time calculation of optimal tracking sequences for arbitrary weather conditions. The tracking strategy is based on simulating all possible tilt configurations for all given weather parameters. It is therefore possible to consider whole weather time series to compute one optimal concise tilt sequence to maximize irradiation while reducing the amount of tilting. An overview of the pipeline can be seen in Figure 1.

The core of dynamic tracking is a machine learning surrogate model for highly spatially resolved irradiance prediction. The increased speed allows us to predict hundreds of tilting scenarios in real time. By training the model on ray tracer generated data, it reproduces the accuracy needed for diffuse tracking while keeping the speed that is inherent to neural networks.

From these scenarios we choose a tracking sequence by utilizing discrete dynamic programming [16]. In this framework all possible tilt sequences are considered, allowing for the development of a truly optimal strategy with respect to average irradiation.

2 | Methodology

2.1 | Surrogate Model

We propose a surrogate model that uses spatial information and weather data to predict the irradiance distribution in a given PV

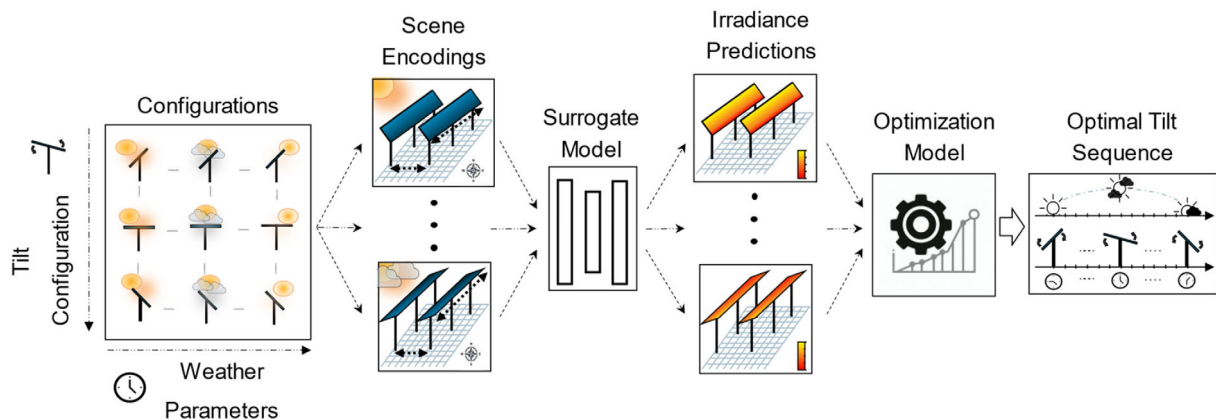


FIGURE 1 | Overview of the dynamic tracking pipeline. All possible tilt configurations for the weather parameters of a given time series are simulated. Then the scene encodings are fed as 2D maps into the surrogate model, which produces spatial irradiance predictions for every scene. These are then passed to the optimization model which computes the optimal tilt sequence.

power plant. This model is trained on simulated data generated with the python package bifacial radiance [17], based on the ray tracer Radiance [18].

As Kurumundayil et al. [15]. showed, encoding spatial information usable for machine learning models is nontrivial but achievable with an appropriate data format. When predicting spatially resolved maps from global parameters, machine learning models tend to lose accuracy. By encoding the spatial information in the shape of an image, the network described in Section 2.1.1 learns to predict the spatial irradiance. As proposed by Kurumundayil, the model takes a scene encoding in the form of a 5-channel input image encapsulating spatial and weather information as input as described in Section 2.1.2. It then generates a spatially resolved irradiance map, which is aligned with the input geometry provided by the scene encoding. In this work, the irradiance values are predicted on module level to estimate the PV energy yield.

2.1.1 | Network Architecture

The underlying architecture of the surrogate model is built upon a U-Net [19] structure. Due to its encoder-decoder structure, the U-Net is adapted to generating image outputs that preserve input semantics while encoding context of the surrounding area at every pixel. The network uses a four-level encoder, each level consisting of two 3×3 convolutions with Leaky ReLU activations followed by 2×2 max-pooling, and a symmetric decoder of four upsampling stages, each with two 3×3 convolutions with Leaky ReLU followed by a 2×2 transposed convolution.

By incorporating dilation [20] into these convolutions, the receptive field is enlarged crucial for modeling long-range shading effects on high pitch panel layouts without inflating the network's number of parameters or losing geometrical information.

As in the regular U-Net structure, three skip connections carry details from encoder to decoder, ensuring structural similarity of

the input and output images. The overall depth is constrained to fit on a standard GPU while still enabling real-time inference.

2.1.2 | Scene Geometry Encoding

The data generation pipeline is shown in Figure 2. Radiance is used to create a scene object from the weather and spatial parameters. To encode the object into an image format, for every individual pixel a sensor is placed in the air facing downward, measuring the irradiance at the corresponding position. The resulting values are assembled into an image. Simultaneously, the individual sensor distances to the spot of measurement are calculated. These distances are assembled accordingly to compose a height map, aligned with the irradiance map. Because every pixel contains the height of the scene at this position, this data representation encapsulates spatial information like tilt, pitch, orientation, and number of rows of the simulated PV array.

For faster execution, these height maps are precomputed. As the spatial properties of the PV plant do not change over time, for one PV plant they only rely on the tilt configuration. To simulate the same power plant multiple times, for different weather scenarios, the height maps are precomputed and stored.

To create an input image for the surrogate model, the weather and meteorological parameters must be encoded. To store these in the same format, the values for DNI, DHI, sun altitude, and sun azimuth are stacked into four additional channels of the input image, resulting in a 5-channel image as input to the surrogate model.

2.1.3 | Dataset Sampling

We want our network to learn various scene configurations, consisting of different geometrical parameters like pitch, tilt, and rotations, as well as weather and meteorological parameters. To sample meteorological data efficiently while keeping realistic combinations of weather parameters, this work follows Kurumundayil et al. [15] in the following way: historical weather data from 2010 to 2019 consisting of DHI, DNI, sun altitude, and azimuth are grouped by months. For every month,

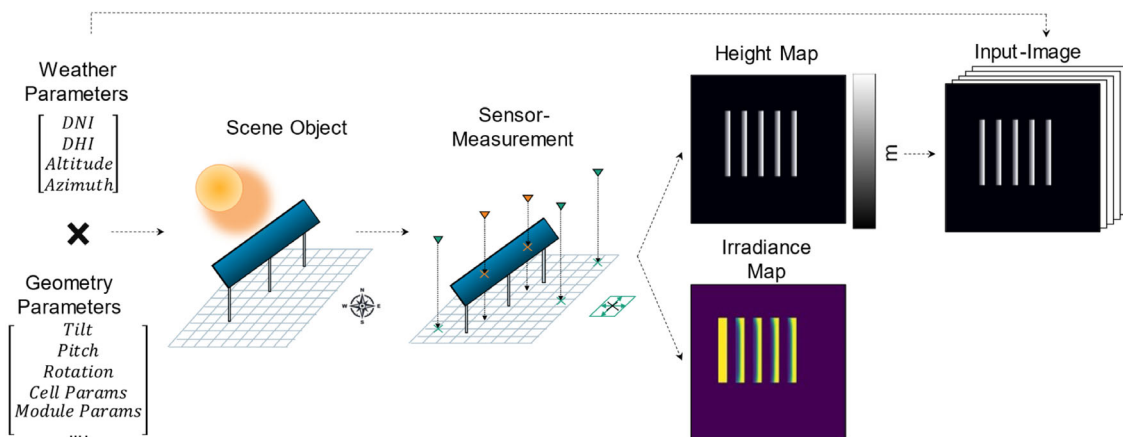


FIGURE 2 | Data generation pipeline. A scene object is constructed from the weather and geometry parameters. The measurements from the sensors are assembled into a grid to form an irradiance map. Simultaneously a height map of the scene is constructed, which is stacked with the weather parameters to form an input image for the surrogate model.

we extract 40 representative weather configurations using the k -means clustering algorithm [21]. The centroids of these clusters are chosen as weather parameters to get 12 times 40 equals 480 configurations. The measurements were sampled from a measurement station in Cologne, Germany, and provided by Deutscher Wetterdienst [22].

To create a spatially diverse dataset, the feasible tilts are discretized into 4° increments across their $[-60^\circ, 60^\circ]$ range. For every tilt increment and for each of the 480 weather configurations, we uniformly sample a pitch from the interval [6 m, 12 m] and east-west rotational deviation from $[-10^\circ, +10^\circ]$. Also, to further diversify the dataset, for every sample, a small random perturbation between 0° and 4° is added to each base tilt value. This yields a dataset of roughly 14 400 samples with a spatial resolution of 512×512 pixels spanning a wide variety of scene geometries and meteorological scenarios.

2.1.4 | Training

The full dataset consists of weather data from 2010 to 2024. While the training set is sampled from historical data from 2010 to 2019, the models are evaluated on the weather data from 2020 to 2024. We randomly split the training dataset of 14 400 samples into 12 960 samples for training (90%) and 1440 samples for validation (10%). The surrogate model was trained for 100 epochs with a batch size of three, using the Adam optimizer [23] (initial learning rate: 10^{-4}) and pixelwise absolute value loss between the network's irradiance prediction and the Radiance ground truth. Training was performed on a Nvidia Titan Xp with a memory of 12 GB.

2.2 | Dynamic Tracking

2.2.1 | Optimization Problem

The goal is to search for a tracking sequence to maximize plane of array irradiation over the course of any given day while restricting ourselves to technical limitations of the tracker and the cost of tilting. This goal is formalized in Equation (1).

We choose a tracking sequence $(T_1^*, \dots, T_n^*)$ for a given time sequence with n weather readings by optimizing the following formula:

$$(T_1^*, \dots, T_n^*) = \arg \max_{T_1, \dots, T_n \in [T_{\min}, T_{\max}]} \sum_{k=1}^{n-1} \overline{f_k(T_k)} \quad (1)$$

$$|T_k| \leq |T_k^b|$$

$$-c(|T_{k+1} - T_k|) + \overline{f_n(T_n)}$$

where the values $\overline{f_k(T_k)}$ are the average irradiance values predicted by our model f at timestep k with tilt T_k . The value $c(d)$ represents the cost of moving the panels d degrees, like motor consumption or an incentive value to decrease too frequent oscillation.

In this modeling framework, we impose the following assumptions. Given that the backtracking tilt is chosen to maximize direct light capture, accounting for diffuse irradiance always reduces the optimal tilt toward zero relative to backtracking. Therefore, the optimal tilt is limited to the backtracking tilt T_k^b in absolute terms as well as to a tracking-system-dependent tilt between $T_{\min}$ and $T_{\max}$. To simulate angular motor precision, tilts are restricted to integer values. Lastly, between two timesteps constant weather parameters are assumed.

2.2.2 | Discrete Dynamic Programming

To solve the optimal control problem from Section 2.2.1, we use discrete dynamic programming [16]. The main idea of discrete dynamic programming is to discretize the state-space, i.e., time series and possible tilts. Therefore, it is possible to simulate all possible tilts for every time step, which allows us to consider all possible tilt sequences. By going backward through all feasible tilts for every time step, we calculate individually, which choice would lead to the best overall result from the current state on out. The best choice is recorded at each step. While iterating forward through the matrix would result in a vast number of possible paths, going in reverse only leads to one optimal path per state, making the optimization feasible. Afterward, we follow those recorded sequences forward in order and because each sequence was picked to maximize the future profit, the resulting sequence is guaranteed to be truly optimal.

The optimization process is depicted in Figure 3. First, all the feasible tilt configurations for every timestep are evaluated individually and assembled into a matrix, which will be called profit matrix. We then traverse backward through the matrix and calculate the profit of every path, also subtracting the cost of tilting.

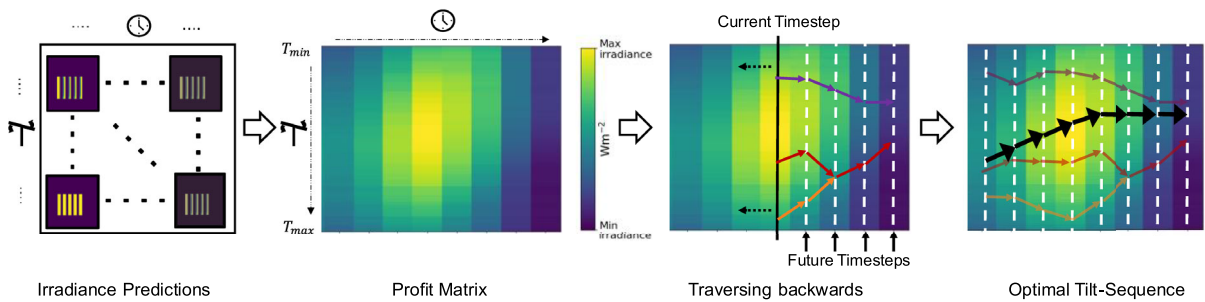


FIGURE 3 | Optimization via discrete dynamic programming. First, all possible tilt configurations and weather combinations are evaluated and assembled into a profit matrix. Then all possible tilt sequences are evaluated by traversing backwards through the matrix, saving the future optimal paths for every state (colored arrows). The optimal tilt sequence is chosen as the one maximizing the profit (black arrows).

Once we have iterated over all time steps, the optimal starting index is chosen as the one with the most total profit. The results of this technique are discussed in Section 3.2.

2.2.3 | Cost and Profit Functions

The adaptability of the cost and profit function of the discrete dynamic programming framework allows for development of different strategies. In this work, the cost function quantifies the motor's energy expenditure for each tilt operation. It is also used to enforce the motor's angular-velocity limit by assigning an infinite cost whenever the proposed change in tilts should exceed it.

To compare irradiance with motor consumption, irradiance must be quantified in terms of power. At present, we use a simplified profit model that computes power gain by assuming a uniform module efficiency of 20% of the incoming irradiance, making it comparable with the tilting cost. Future work will integrate more advanced PV models to capture spatial, angular or temperature effects as discussed in Section 4.2.

3 | Experimental and Results

3.1 | Validation of the Surrogate Model

3.1.1 | Evaluation Approach

The quality of the surrogate model is tested by generating spatially resolved predictions and comparing them with the irradiance maps produced by the ray tracer. The model was trained with the data described in Section 2.1.4. We show the applicability of the model for weather data after 2020, which were not used during training and validation of the model. Each test case

simulates a real-world five-row PV array of forty modules per row, with a 6 m pitch and east–west tracking capability, thereby standardizing the plant geometry across all simulations. The proof of spatial generalizability of this model for varying scene geometries is part of a future work and is discussed in Section 4.1.

Since we have the large-scale application in mind, for temporal validation we select the center PV row for evaluation because it is closest to the repeating boundary conditions found in a multirow PV power plant.

We choose a resolution of 512×512 pixels for a space of 55×55 m, resulting in one pixel per 10.7 cm, as a trade-off between spatial accuracy and size of the images.

3.1.2 | Results on Spatial Irradiance Prediction

We give two spatially resolved examples of different tilts and weather scenarios. For both scenarios, the area beneath the modules was masked by applying a height-based threshold. The results can be seen in Figure 4.

The surrogate model accurately predicts the spatial irradiation distribution in both scenarios. In scenario 1, the pixelwise average of the predicted irradiance is calculated for hourly readings over the course of a day with almost no direct irradiance. We reach a maximal deviation of 12.31 Wm^{-2} and an average deviation of 1.43 Wm^{-2} , which corresponds to 0.78% of the average DHI of 182.27 Wm^{-2} . In scenario 2, we predict the irradiance for a sunny morning with a DNI of 830 Wm^{-2} and a DHI of 116.38 Wm^{-2} for a sun with an altitude of 45.4° , azimuth of 11.5° , and a PV module tilt of 45° to the east.

The model accurately predicts all but the easternmost PV string to be shaded. The maximal deviation of 46.12 Wm^{-2}

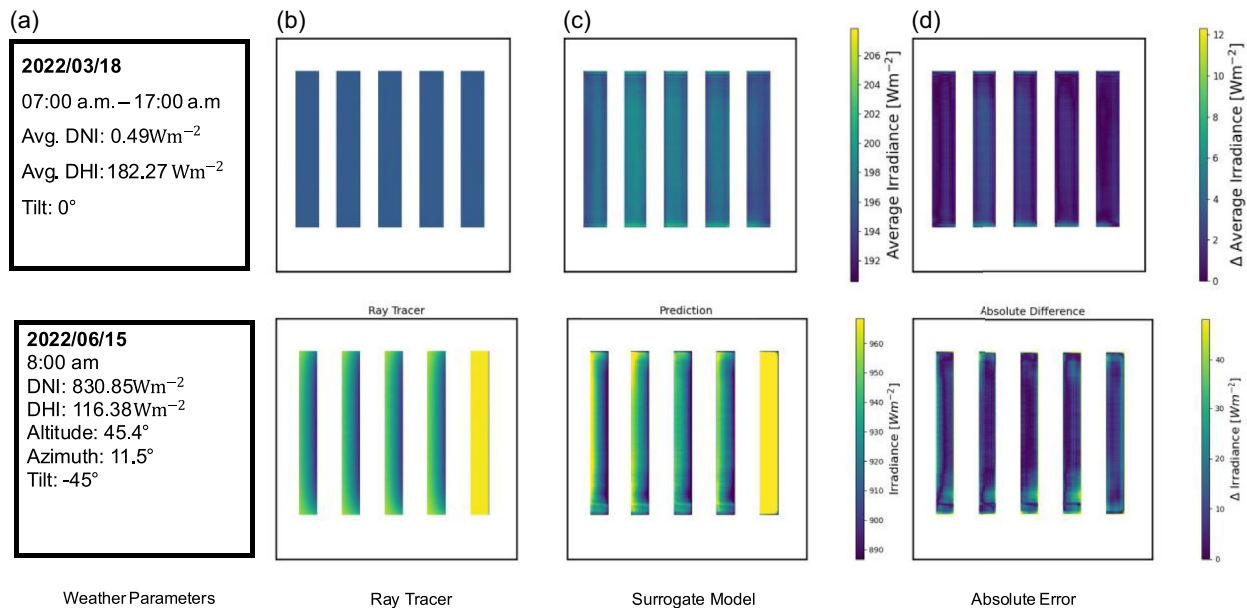


FIGURE 4 | Comparison of ray tracer to surrogate model output by absolute distance. Row 1 shows the average irradiance for hourly predictions of a horizontally tilted PV system on a cloudy day in March 2022. Row 2 shows the irradiance for a PV system tilted to 45° east on a sunny morning in June 2022. The resolution was increased for visibility.

or 6.5% of the global horizontal irradiance (GHI) occurs at the southern edge, where the surrogate model predicts a slightly deeper shadow than the ray tracer. An average deviation of 8.97 Wm^{-2} is achieved, which corresponds to 1.3% of the GHI.

3.1.3 | Results on Irradiance Prediction for Varying Configurations

We analyze the performance of our surrogate model in two steps: first, we show the effect of varying pitches and tilt configurations over the course of 1 day. The results can be seen in Figure 5. To show the model's performance on days with high irradiance, this day was chosen as the 95th percentile in terms of average DNI in 2022. The plot shows the mean relative error calculated as mean value of the pixelwise error on the modules for tilts between -60° and 60° for pitches of 6, 9, and 12 m.

The model aligns with the ray tracing outputs staying mostly within a mean relative error of between -3% and 3% . For

situations with a very low solar altitude and little irradiance ($5:00\text{--}6:00$, $19:00\text{--}20:00$), the model struggles with tilts close to the backtracking tilt. The outputs align through varying tilts and pitches, showing generalizability for different power plant configurations.

In a second step, we validate our model for a whole year with varying tilts for a power plant with a fixed pitch of 9 m. The results can be seen in Figure 6. For this evaluation, we simulated each day in 2022 as test cases. Timestamps outside of backtracking were masked. For every tilt and hour, the model outputs were averaged over the corresponding month ignoring the masked samples. The plot depicts the mean relative error.

In weather situations with a high solar elevation, our model closely aligns to the testing-data. Near sunset, especially, when DNI and DHI are low, errors increase. Across all unmasked samples, the average relative error is 3.1% with a mean absolute error of 9.2 Wm^{-2} .

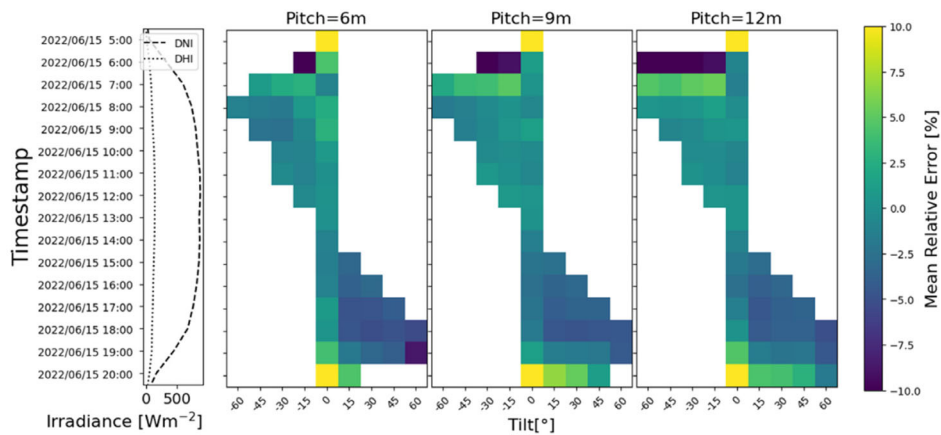


FIGURE 5 | Weather parameters and mean relative errors of surrogate model and ray tracing output. Timestamps outside of backtracking were masked. .

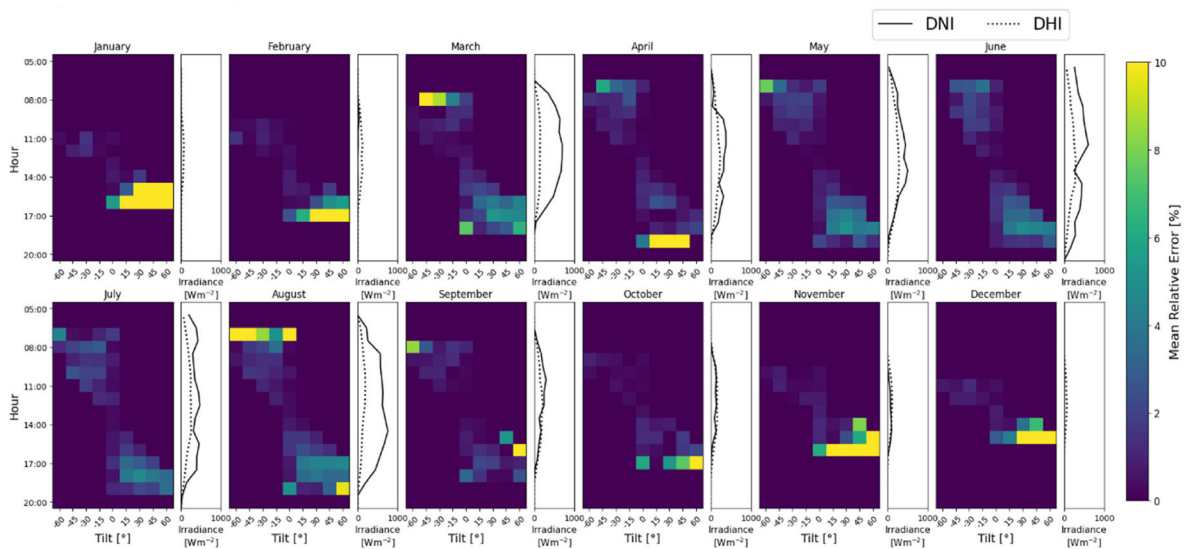


FIGURE 6 | Absolute relative error of the surrogate model for different tilt and weather configurations. Timestamps outside of backtracking as well as time stamps with a solar altitude below zero were masked. Every second column shows the weather parameters aligned in time.

3.2 | Application: Spatial Irradiance Analysis via the Surrogate Model

We show the benefit of the spatial resolution of the surrogate model by comparing the irradiance distribution of the backtracking tilt and the dynamic tracking tilt for overcast scenarios.

Figure 7 presents the irradiance distribution across all evaluated tilt configurations for one timestep of an overcast day (22/03/18, 11:00). Each bar corresponds to one tilt configuration, with its height indicating the average irradiance received under that configuration. To enhance the visibility of spatial inhomogeneity, the fraction of irradiance values falling substantially below the maximum for each configuration is binned and color-coded.

The dynamic tracking tilt achieves a higher average irradiance and greater homogeneity than the backtracking tilt. This difference and the slope of this curve are attributed to the dominance of diffuse irradiance under overcast conditions. Therefore, configurations with a larger tilt experience inhomogeneity due to diffuse shading. Dynamic tracking achieves an increased averaged irradiance of 239.6 Wm^{-2} opposed to 210.0 Wm^{-2} for backtracking. Due to less shading, it also exhibits a more homogeneous irradiance distribution compared to the backtracking configuration as indicated by the monochrome bar.

This plot also shows the benefits of using an anisotropic radiation model in Radiance. [24] While isotropic radiation models expect diffuse irradiation to arrive from all sides equally, anisotropic models consider circumsolar and horizon brightening effects. A model trained on solely isotropic data would therefore predict a slope that is centered around a tilt of 0° , while the one displayed here is shifted due to the sun's position at 11 a.m.

3.3 | Dynamic Tracking

The significantly reduced prediction time enables us to simulate all possible tilt sequences to develop a new method to efficiently search for an optimal tilt sequence throughout the course of a whole day. We call this framework dynamic tracking due to its discrete dynamic programming backbone.

3.3.1 | Experimental Setup

We simulate a PV power plant consisting of 5 rows of PV strings and evaluate the middle row to include the effects of self-shading while emulating the repeating condition of a larger power plant. Empirical findings show that the energy used for the motors in the real testing area is approximately 0.018 Whm^{-2} per degree of movement per row. The cost function also allows us to limit the maximum tilt angle between two timesteps to the maximum angular velocity of the simulated motor of 9° per minute by setting the cost to infinity if exceeded. To align consumption with a module efficiency of approximately 20% of the irradiation, we scale the cost values accordingly.

3.3.2 | Dynamic Tracking Strategy

We test our optimization method using historical weather data from 2 days in September 2024 with varying weather conditions. Therefore, the discrete dynamic programming routine is applied, as explained in Section 2.2. Figure 8 presents the optimal tracking sequence for these 2 days.

The optimization model shows features of backtracking as well as horizontal tilting strategies. In scenarios with a low diffuse ratio (row 1, 10:00–13:00), the model closely adheres to the backtracking algorithm. Conversely, in times with a higher diffuse ratio

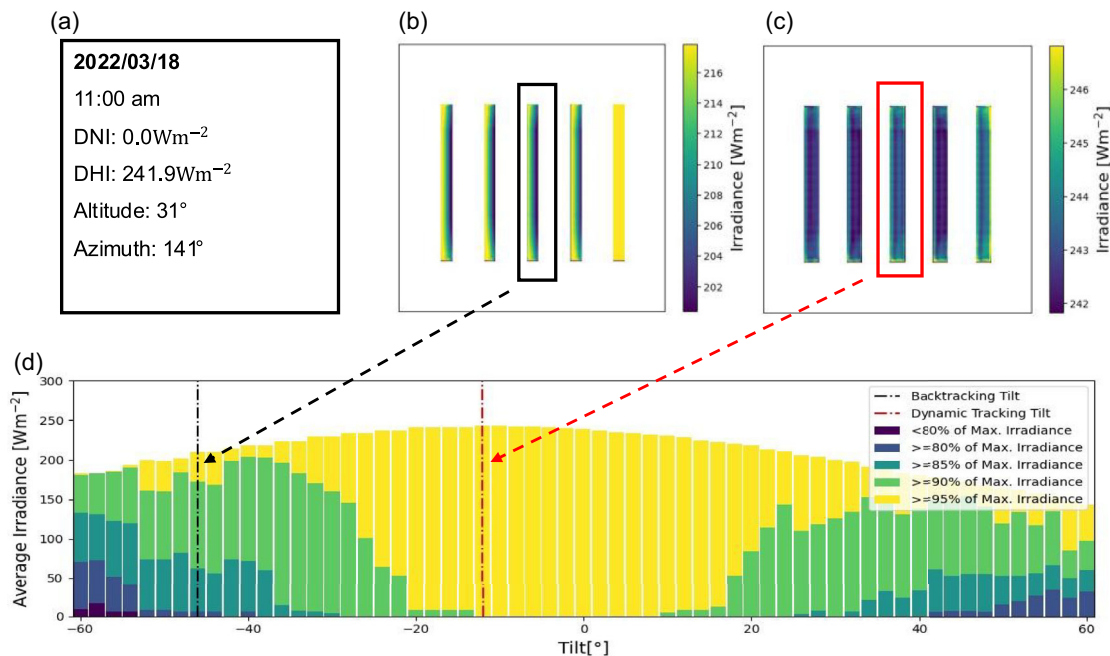


FIGURE 7 | (a) Comparison of irradiance distribution for an overcast scenario of (b) backtracking tilt and (c) dynamic tracking tilt. Row 1 shows the spatial distribution of the irradiance prediction of our surrogate model. Row 2 shows the value as well as binning of the irradiance (d). Every bar resembles one tilt configuration. The height of the bar depicts the average irradiance; the colors show the variation of irradiance.

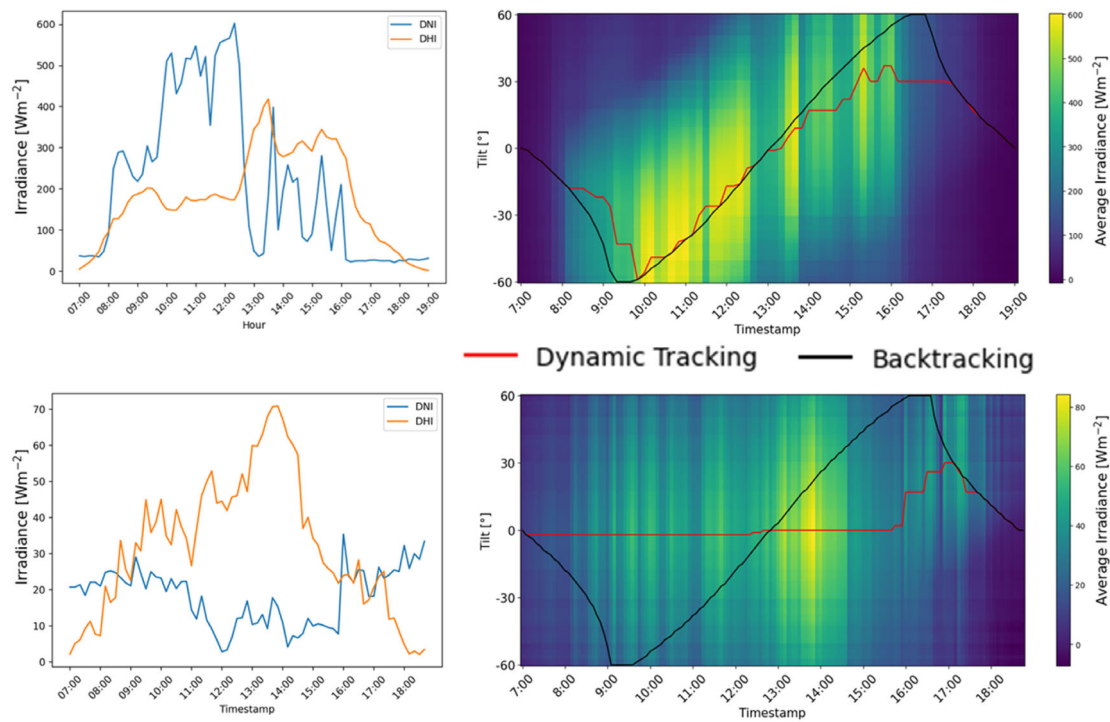


FIGURE 8 | The optimal tracking sequence for 2 days in September 2024. Top: A day with varying level of diffuse ratio (24/09/22). Bottom: A cloudy day (24/09/26). Column 1 shows the DNI and DHI values over the whole day in averaged timesteps of 10 min. Column 2 shows the backtracking path (black) as well as the dynamic tracking path (red).

(row 2, 07:00–15:00), the model deviates from this strategy by staying closer to zero tilt. Furthermore, during periods with fluctuating diffuse ratios (row 1, 15:00–17:00), dynamic tracking employs a mixed strategy, demonstrating resilience to changing weather conditions, likely due to the costs associated with tilting.

In the top scenario, an average increase in irradiation of 2% compared to backtracking and 16% compared to a fixed tilt of zero degrees was achieved. In the bottom scenario, the average irradiation increased by 22% over the backtracking and 4% over the zero-tilt.

3.3.3 | Computational Performance

The calculations for the top scenario Figure 8 involved 8833 predictions, derived from 73 timestamps multiplied by 121 possible tilts resulting from a limit of -60 to 60° and a rotational precision of 1° . The calculations, including optimization, were completed in approximately 10 min on a consumer graphics card. By restricting the possible tilts to those limited by backtracking, the number of computations is reduced by 80% resulting in a computation time below 2 min.

3.3.4 | Dynamic Tracking Gain

We demonstrate the potential of dynamic tracking by optimizing PV tracking over the course of the entire year 2022. Figure 9 presents the simulated gain of dynamic tracking compared to conventional backtracking. The results were smoothed using a 10 day averaging window. By construction, dynamic tracking consistently outperforms backtracking, with the highest gain

in months characterized by a high diffuse ratio (e.g., January and December).

Over the full year, dynamic tracking yields average irradiation gains of 2.93% for a pitch of 6 m, 3.73% for a pitch of 9 m, and 4.45% for a pitch of 12 m. The maximum relative improvement is observed in December, reaching 34.77% for a pitch of 6 m, 21.19% for a pitch of 9 m, and 21.09% for a pitch of 12 m.

4 | Discussion and Outlook

4.1 | Surrogate Model

The surrogate model has proven to be generalizable across different weather and tilt conditions, matching ray tracer predictions. While equivalent performance on PV plants with different layouts, such as orientation or height is expected, the demonstration of spatial generalization will be part of a separate work. For out of distribution predictions like novel layouts for PV power plants, the surrogate model can be retrained with additional samples. To improve the accuracy of our model regarding low solar altitude, we can adapt our training set to include more of these data points or increase variety by adding noise to our training weather parameters.

The surrogate model reliably predicts self-shading arising scenarios. To increase prediction accuracy further, the training process can be adapted. While convolutional neural networks (CNNs) excel in terms of computational speed and at predicting diffuse shading patterns, they have architecture-related limitations. CNNs tend to blur the prediction of high-contrast semantic

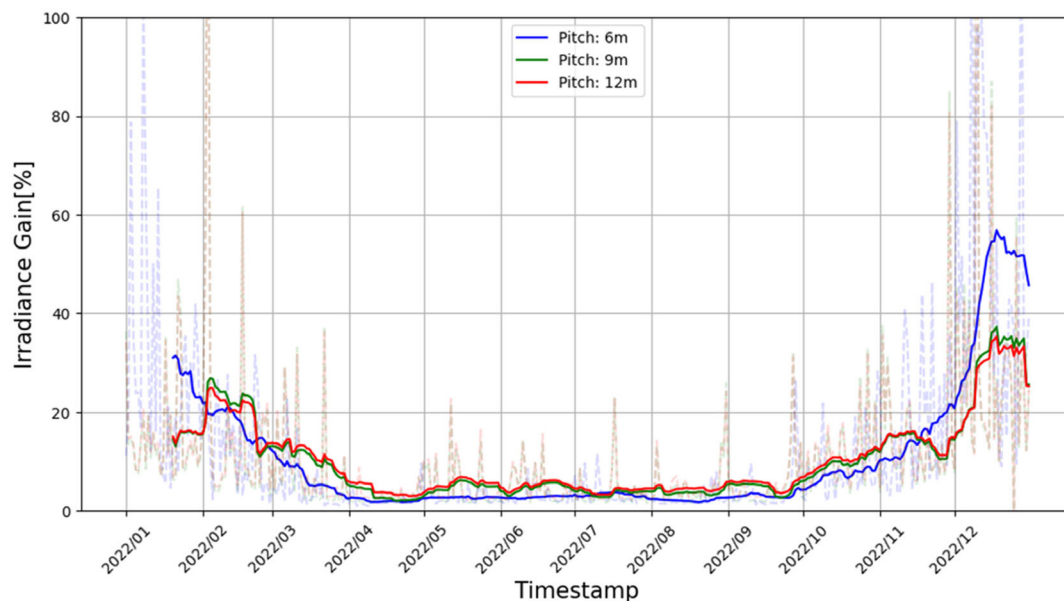


FIGURE 9 | Gain in POA irradiance of dynamic tracking against backtracking over the course of 2022. Pitches with a length of 6, 9, and 12 m are compared. The values were averaged with a window size of 10 days.

edges, such as deep shadows. This smoothing effect is inherent to training with pixel-wise losses like the mean squared error or mean absolute error, which penalize outliers and bias the network toward average-value reconstructions. Future work could mitigate these effects by varying the training process using adversarial trainings such as GANs [15] or perceptual losses in some stage of the training process [25].

4.2 | Tracking Optimization

Comparative evaluation of dynamic tracking against conventional rule-based controllers is nontrivial because most established strategies rely on discrete case distinctions. By contrast, dynamic tracking formulates one single optimization over all configurations, eliminating the need for manually defined thresholds. Although threshold-based heuristics can be reproduced within the same framework, the surrogate-driven dynamic programming approach directly computes the true optimal tilt sequence, enabling unbiased performance assessment and straightforward adaptation to any PV power plant.

As PV power conversions are highly nonlinearly dependent on irradiance, this framework can greatly benefit from implementing different PV models. In the current model, the profit of the PV modules is computed by applying a fixed module efficiency to irradiance and subtracting a simple motor-energy cost. Future work will therefore cover models that incorporate spatial effects or inverter performance.

Currently, dynamic tracking assumes perfect knowledge of future DHI and DNI over the optimization horizon. For real-time deployment, the optimizer could be paired with short-term irradiance forecasting and embed these forecasts in a receding-horizon control loop [26–28]. This approach enables adaptive tilt updates and explicit treatment of forecast uncertainty, improving robustness under changing sky conditions.

5 | Conclusion

We presented a novel framework for tracking optimization applicable under arbitrary weather conditions, accounting for diffuse and direct irradiation.

Through the integration of a machine learning surrogate model for irradiance prediction and the combination with discrete dynamic programming, this framework can optimize PV tracking sequences over entire days within less than 2 min. Due to the fast computation of a full cost matrix with varying tilt for given weather conditions, the dynamic tracking approach identifies the optimum tracking sequence. Even the costs of motor consumption can be considered.

We demonstrated the utility of the surrogate model by analyzing the front side irradiance distribution of the PV array, showing that dynamic tracking achieves greater homogeneity than conventional methods. For the mixed weather scenarios shown, this strategy outperforms existing strategies, increasing irradiation consistently by up to 34.5% over backtracking and zero-tilt strategies.

The surrogate model was validated in various scenarios, including different tilt configurations as well as different weather parameters. The scene encoding allows the application of the U-Net architecture for fast and reliable irradiance prediction.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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